

Writing parameterized modules

Proposed problems

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Problem 1

Construct a parameterized 4-to-1 multiplexer, named *mux2s*. The module should be parameterizable by its width, having the following interface:

```
1 module mux2s #(
2     parameter w = 4           //width parameter
3 )(
4     input  [w-1:0] d0 ,d1 ,d2 ,d3 , //4 data inputs
5     input  [1:0] s ,             //selection input
6     output [w-1:0] o             //data output
7 );
```

Build a testbench for verifying the multiplexer. In the testbench, the parameter *w* will have a value of 8 and the multiplexer should be tested with at least 8 distinct input configuration.

Note: Use the \$urandom() Verilog system call for generating 32 bits of random data.

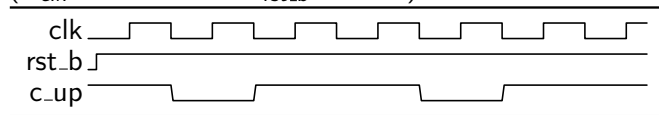
Problem 2

Construct a synchronous counter, named *count11*. The module should be parameterizable with its width and initial value. The counter's interface, besides *clk*, includes the following signals:

- *rst_b*, asynch., active low, sets content to *initial value*
- *c_up*, synch., active high, increments counter's value by 11
- *q*, output, counter's content

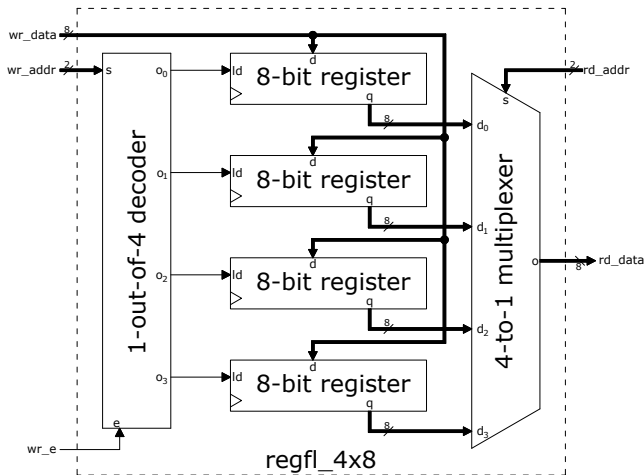
Parameterize the counter on 10 bits with an initial value of 'h3fb and test it with the inputs given in the diagram bellow

($T_{clk} = 200ns$, $Pulse_{rst_b} = 10ns$):



Problem 3

Construct the 4x8 register file architecture bellow



The clock and reset lines were omitted for clarity.

Note: The Verilog code of a width-parameterizable, parallel load register is available

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Problem 3 (contd.)

The 4x8 register file will have the following interface

```
1 module regfl_4x8 (  
2     input  clk ,  
3     input  rst_b , //asynch  
4     input  [7:0]  wr_data ,  
5     input  [1:0]  wr_addr ,  
6     input  wr_e ,  
7     output [7:0]  rd_data ,  
8     input  [1:0]  rd_addr  
9 );
```

The register file has no read enable input, *rd_e*, meaning that at any moment, one of the 4 internal registers' content will be delivered to *rd_data*.

Problem 3 (contd.)

Test the 4×8 register file with a testbench that generates the inputs as in the diagram bellow ($T_{clk} = 100ns$, $Pulse_{rst_b} = 5ns$)

